

Bank of Namibia Data Engineering Technical Framework

Purpose

The purpose of this framework is to outline the current layout and the technological landscapes of the Data Analytics infrastructure at the Bank of Namibia, segmented into two components, Cloud-based, and On-Premises based.

Target Audience

This framework is intended for Bank of Namibia IT personnel as well as any stakeholders who require its use for ease of reference in line with the Digital Transformation programme of the Bank.

Context

The Bank of Namibia uses various technology tools anchored on this framework to manage, transform, and deliver data efficiently and reliably for various use cases in the Bank.

Data Storage and Processing

The Bank of Namibia stores and processes large volumes of data from various sources and formats. Most of this data is housed in various systems.

Types of data storage and processing systems used:

- Relational databases, data warehouses, data lakes.
 - Microsoft SQL Server/Azure SQL Database for structured data.
 - Microsoft SQL Server Integration Services and Azure Synapse Analytics for data warehousing.
 - Azure Data Lake for data lake storage.

Data Integration and ETL/ELT

We integrate and transform data from various sources and formats into a consistent and usable form for analysis and consumption.

ETL (extract-transform-load) or ELT (extract-load-transform) tools and frameworks that automate and streamline the data integration process is used.

Toolsets:

- Microsoft SQL Server/Azure SQL Database
- Data Pipelines (SQL Server Integration Services/Azure Data Factory)
 - Connecting and loading data from different sources
 - Orchestrating and scheduling data pipelines
- Data Warehouse (SQL Server Integration Services/ Azure Synapse Analytics)
 - Transforming and manipulating data
- Tabular/Dimensional Cubes (SQL Server Analysis Services/Azure Synapse Analytics)

Data Quality and Testing

To ensure the quality and reliability of the data and pipelines:

- Check and validate the data for accuracy, completeness, consistency, timeliness, and compliance.
- Test and monitor the pipelines for performance, availability, and error handling.
- Define and verify data quality expectations.
- Collect and visualize pipeline metrics and alerts.
- Manage and troubleshoot pipeline failures.

Data Modelling and Design

Model and design data structures and schemas that support our data analysis and business goals. The model must answer business-level questions independently of the implementation:

- Choose the appropriate data modelling techniques and patterns.
 - Dimensional modelling, star schema, snowflake schema, or data vault, according to data characteristics, requirements, and use cases.
- Apply best practices and principles of data modelling,
 - Normalization, denormalization, referential integrity, or data governance.
- Create and document the data models.
- Define and generate the data schemas.
- Manage and cataloguing the data assets.

Data Analysis and Visualization (Microsoft Power BI)

To enable and facilitate data analysis and visualization for our data consumers and stakeholders.

- Provide the necessary data access and security controls.
- Authentication, authorization, encryption, or auditing.
- Select and integrate suitable data analysis and visualization tools and frameworks.
- Interactive data exploration and prototyping.
- Collaborative data analysis and sharing.
- Microsoft Power BI for creating and delivering data dashboards and reports.
 - Power BI assets:
 - Dashboards, Reports and KPI's
 - Datasets